

# From Error Curves to an Estimated Weight: A Model-Based Reanalysis of Perspective Taking Across Adulthood

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## Abstract

Egocentric errors in the director task become more common across later adulthood, but descriptive error curves do not identify how the underlying interpretive process changes. We reanalyzed the open behavioral data of Bradford et al. (2023) using a choice-level projection of Heller et al.’s (2016) probabilistic mixture model — in which a weight  $\alpha$  indexes the egocentric contribution to interpretation — with a Beta–Binomial participant hierarchy and a publicly frozen robustness protocol. The age-dependent model estimated a contrast of  $\Delta\alpha = +0.127$  between ages 25 and 75, with a 95% participant-bootstrap interval of  $[+0.062, +0.196]$ . It also predicted entirely held-out participants completing the same task and items better than the fitted constant- $\alpha$  model in all 20 cross-validation repetitions. However, a prespecified parametric predictive check failed among younger adults: 18 participants aged 20–37 made exactly one egocentric error, exceeding the model’s 99% predictive interval of  $[1, 11]$ . Thus, an age-varying mixture weight provides a robust and predictively useful description of behavioral differences across adulthood, but a single mixture process does not reproduce the full distribution of observed errors. Because final choices cannot distinguish probabilistic weighting from an equivalent lapse process, all psychological interpretations of  $\alpha$  remain conditional on the assumed response rule.

**Keywords:** perspective-taking; common ground; reference resolution; cognitive aging; computational modeling; director task

# 1 Introduction

Understanding another person often requires setting aside information that is obvious from one’s own perspective (Samuel et al., 2026). The problem is clearest in reference resolution, where a listener must work out which object a speaker means. Suppose a speaker and a listener both see two candles, and a third, smaller candle is visible only to the listener. When the speaker asks for “the small candle,” the candle that is smallest in the listener’s view cannot be the intended referent — the speaker does not know it exists. To interpret the request correctly, the listener must weigh not only which object best fits the description, but which objects the speaker can see at all. Yet listeners sometimes look at, and occasionally reach for, the hidden candle anyway (Keysar et al., 2000) — the classic egocentric error.

The processes underlying such errors — documented most extensively in the “director task” paradigm (Keysar et al., 2000) — remain disputed. Egocentric-anchoring accounts propose that listeners initially interpret an utterance from their own perspective and only later, through an effortful correction process, adjust toward the speaker’s knowledge; errors arise when that adjustment is incomplete or delayed (Keysar et al., 2000; Epley et al., 2004). Early-integration accounts instead argue that common-ground information constrains reference resolution from the earliest moments of processing, such that listeners do not necessarily begin from a fully egocentric interpretation (Hanna et al., 2003). A third possibility is that listeners represent both egocentric and common-ground interpretations concurrently, with behavior reflecting their relative probabilistic weighting rather than a fixed sequence of anchoring and correction (Heller et al., 2016). These accounts can produce similar final choices even though they imply different underlying computations, making aggregate error rates insufficient for deciding among them. More fundamentally, the director task itself has been criticized as potentially confounding perspective-taking with selective attention and task demands, leaving open whether an “egocentric error” uniquely identifies a failure to use common ground at all (Rubio-Fernández, 2017).

Existing computational work has formalized perspective use, but has not examined how a comprehension-side perspective-mixture weight varies across adult participants in the director task. Heller et al. (2016) introduced a probabilistic mixture parameter governing the relative contribution of egocentric and common-ground information, but used it to generate qualitative predictions

rather than estimating its value from data. Mozuraitis et al. (2018) subsequently examined which ranges of a related parameter reproduced condition-level patterns in reference production, without trial-level likelihood estimation, participant heterogeneity, or age dependence. Other neighboring approaches have modeled common ground as an inferred latent state evaluated against offline judgments (Rubio-Fernández and Jara-Ettinger, 2018) or fitted Rational Speech Act parameters in a different communicative paradigm (Hawkins et al., 2021). To our knowledge, no previous study has estimated a comprehension-side perspective-mixture weight from director-task choices using a participant-level hierarchy, modeled that weight across adulthood, and evaluated the resulting age-dependent account through held-out-participant prediction.

Bradford et al. (2023) provide the empirical basis for the present analysis. They studied a community lifespan sample of adults aged 20–86 (analytic  $N = 264$ ) using an eye-tracked, computerized version of the director task, in which each participant completed 12 critical Listener-Only trials requiring them to ignore a privileged competitor and interpret the director’s intended referent. Their behavioral analysis found a quadratic relation between age and egocentric error rates: performance remained broadly stable through approximately age 37, followed by a substantial increase in egocentric errors from around age 38 onward. The study also recorded eye movements and made both the data and analysis code openly available, creating an unusually rich basis for computational reanalysis. However, their analysis characterized how observed error rates changed across adulthood; it did not estimate the parameters of a cognitive model intended to explain those choices.

The present study fitted a choice-level projection of the mixture model of Heller et al. (2016) to the director-task choices of Bradford et al. (2023), using a Beta–Binomial participant hierarchy and a publicly frozen analysis protocol. First, the model quantified how the estimated egocentric contribution varied across adulthood: the contrast between ages 25 and 75 was  $\Delta\alpha = +0.127$ , with a 95% participant-bootstrap interval of  $[+0.062, +0.196]$ . Second, the age-dependent model predicted entirely held-out participants completing the same task and items better than the fitted constant- $\alpha$  model in all 20 cross-validation repetitions. Third, a prespecified parametric predictive check revealed a systematic failure among younger adults, showing that a single age-varying mixture process does not reproduce the full observed distribution of errors. The analysis therefore yields not only an estimate of age-related variation in  $\alpha$ , but also evidence about where that quantitative account succeeds and where it breaks.

These results do not adjudicate among the three process accounts introduced above. At the level of final two-alternative choices, a probability-matching mixture weight is formally indistinguishable from an argmax-plus-lapse process under the mapping  $\varepsilon = 2\alpha$  (Proposition 1), so every estimate is conditional on the assumed response rule; the  $\Delta\alpha$  above reads equally as a lapse-rate change of +0.254. Final choices likewise cannot determine whether competing perspectives were represented simultaneously or arose through an egocentric interpretation followed by correction. What the present study offers instead is one explicit account made quantitative, its predictive value measured in held-out participants, and its empirical failure point located.

## 2 Model

### 2.1 A choice-level projection of the domain-mixture account

Heller et al. (2016) proposed that listeners resolve definite reference by weighing two referential domains simultaneously: an egocentric domain (all objects the listener can see) and a common-ground domain (objects visible to both interlocutors). Their Equation 2 combines Bayesian reference resolution under each domain with a weight  $\alpha$  on the egocentric domain,

$$P(obj \mid RE) = \alpha P(RE \mid obj, d=e) P(obj \mid d=e) + (1 - \alpha) P(RE \mid obj, d=c) P(obj \mid d=c), \quad (1)$$

where an  $\alpha$  near 1 corresponds to egocentric interpretation and  $\alpha$  near 0 to full common-ground restriction. In the original paper,  $\alpha$  was varied from 0 to 1 to generate qualitative predictions; it was not estimated from data.

We fit a deliberately reduced projection of this model to choice data. On a critical director-task trial, the display admits exactly one best egocentric referent (a privileged competitor matching the instruction) and one best common-ground referent (the intended target). With point-mass within-domain resolution, the model reduces, per trial, to

$$P(\text{select competitor}) = \alpha, \quad P(\text{select target}) = 1 - \alpha, \quad (2)$$

under a probability-matching response rule. This projection discards three components of the full

account — graded, production-normed referring-expression likelihoods; a ground-status prior; and incremental (partial-expression) evaluation — and therefore estimates  $\alpha$  only at the level of final choices.

## 2.2 Hierarchy

Participants vary. We model participant  $i$ 's weight as

$$\alpha_i \sim \text{Beta}(\mu(\text{age}_i) \kappa, (1 - \mu(\text{age}_i)) \kappa), \quad e_i \sim \text{Binomial}(n_i, \alpha_i), \quad (3)$$

yielding an exact Beta-Binomial marginal likelihood. The primary age model is logit-quadratic,

$$\text{logit } \mu(\text{age}) = \beta_0 + \beta_1 z + \beta_2 z^2, \quad z = (\text{age} - 53)/10; \quad (4)$$

the comparator is the fitted constant- $\alpha$  model  $(\alpha_0, \kappa)$ . The primary estimand is  $\Delta\alpha = \alpha(75) - \alpha(25)$ .

## 2.3 Response-rule conditionality

**Proposition 1.** *For two-alternative trials with point-mass domains, the probability-matching rule with weight  $\alpha$  and an argmax-plus-lapse rule with lapse rate  $\varepsilon$  (lapses uniform over the two alternatives) induce identical likelihoods for every possible dataset under the mapping  $\varepsilon = 2\alpha$ .*

*Proof.* Under matching,  $P(\text{competitor}) = \alpha$ . Under argmax-plus-lapse with  $\alpha < .5$ , the argmax is the target, so  $P(\text{competitor}) = \varepsilon/2$ . Setting  $\varepsilon = 2\alpha$  equates the trial likelihoods, hence all products of them.  $\square$

Consequently, choice data cannot distinguish “egocentric weighting” from “lapsing” at this level: every  $\alpha$  reported below carries its dual reading  $\varepsilon = 2\alpha$ , and all psychological interpretation is explicitly conditional on the response rule. Process-level data (e.g., graded fixation measures) are required to separate the readings; we return to this in the Discussion.

## 3 Method

### 3.1 Data

We reanalyze the open dataset of Bradford et al. (2023) (<https://osf.io/2epsu/>), an eye-tracked computerized director task with a community lifespan sample. Participants followed prerecorded instructions to move objects in a  $4 \times 4$  grid with occluded slots; on Listener-Only (critical) trials the occluded object matched the instruction (e.g., the smallest of three candles, hidden from the director), so correct interpretation required the director’s perspective; on Shared-Perspective (control) trials the occluded object did not match. Each participant contributed 12 critical and 12 control trials. We model critical trials; the trial-level outcome is the authors’ EgocentricErrors coding (selection of the hidden competitor); other errors ( $\approx 1.2\%$  of trials) count as non-egocentric, so the likelihood estimates egocentric-choice probability specifically.

### 3.2 Cleaning, with verification against published anchors

Exact byte-duplicate rows were removed (48 rows from two participants whose sessions were duplicated in the file); one participant absent from the demographics file and one with FSIQ4  $< 70$  were excluded, reconstructing the paper’s analysis sample of  $N = 264$  (ages 20–86), each with exactly 12 critical trials. Two published anchors were reproduced before any model contact: the analysis  $N$  (264) and the mean per-participant egocentric rate (10.23%, matching the 10.23% reported by Bradford et al., 2023). The cleaning pipeline halts if any expectation fails.

### 3.3 Protocol history

All analysis decisions were frozen in a public, version-controlled protocol before the outcome-bearing run (repository commits `e09d306`, amended `a8de96b` and `be30e90` after two rounds of structured adversarial review elicited by the author from the assisting AI system in separate sessions (see the AI Assistance Disclosure); implementation `63943b1`). The protocol is explicitly a *frozen analysis following exploration*: earlier exploratory passes on the same data (committed and public) informed the model shape and the two predictive-check statistics, and this is disclosed rather than laundered. One notable exploratory correction is retained in the record: an initial fit with a bounded heterogeneity grid overstated the age effect (a  $\kappa$ -floor artifact); freeing  $\kappa$  revised the trajectory downward

and widened the interval. Held-out participant prediction, single-shot execution, and complete reporting carry the evidential weight.

### 3.4 Estimation and evaluation

Maximum likelihood by a deterministic coarse grid followed by five-start bounded Nelder–Mead (tolerance  $10^{-6}$ ; bound-hits halt the run; numpy 2.4.4, scipy 1.17.1). Model comparison by AIC. Predictive evaluation by participant-level 5-fold cross-validation, 20 repeats, folds stratified by fixed age bands, fixed seeds; both models refit within every training set; the prespecified stability rule (positive held-out difference in  $\geq 17/20$  repeats) is descriptive, not an inferential test, because repeated cross-validation estimates are correlated. Uncertainty for  $\Delta\alpha$  by participant-level cluster bootstrap ( $B = 1000$ , fixed generator, percentile intervals). Parametric predictive checks (from maximum-likelihood point estimates; not posterior predictive) used two prespecified statistics: T1, the pooled egocentric rate at ages 73–86; T2, the number of participants aged 20–37 with exactly one egocentric error. Sensitivity analyses: finer grids; IQ-cutoff variants; knee-shaped alternatives with a prespecified identifiability criterion for the knee; full leave-one-participant-out; leave-one-trial-position-out.

### 3.5 Scope

Cross-validation here assesses generalization to unseen participants within the observed task and item set; it does not assess generalization to new items, task variants, populations, or laboratories.

## 4 Results

### 4.1 Primary fit

The primary model estimated  $\beta_0 = -2.423$ ,  $\beta_1 = 0.279$ ,  $\beta_2 = 0.093$ ,  $\kappa = 1.38$  (log-likelihood  $-647.18$ ), against the fitted constant- $\alpha$  model  $\alpha_0 = 0.112$  ( $\varepsilon$ -dual 0.225),  $\kappa = 1.24$  (log-likelihood  $-658.41$ );  $\Delta\text{AIC} = -18.48$ , meeting criterion R1. Table 1 gives the fitted curve values:  $\alpha(25) = 0.078$  [95% CI 0.046, 0.118],  $\alpha(50) = 0.076$  [0.050, 0.104],  $\alpha(75) = 0.205$  [0.147, 0.268], with  $\Delta\alpha = +0.127$  [ $+0.062$ ,  $+0.196$ ] ( $+0.0025/\text{yr}$  averaged). The curve is not monotonic over  $[25, 75]$ : it declines shallowly to a minimum at age 38.0 and rises with acceleration thereafter (0.089 at 20; 0.067 at 40;

Table 1: Primary estimates with bootstrap 95% CIs and response-rule duals. All values conditional on the probability-matching rule (Proposition 1).

| Quantity       | Estimate | 95% CI           | $\varepsilon$ -dual |
|----------------|----------|------------------|---------------------|
| $\alpha(25)$   | 0.078    | [0.046, 0.118]   | 0.155               |
| $\alpha(50)$   | 0.076    | [0.050, 0.104]   | 0.152               |
| $\alpha(75)$   | 0.205    | [0.147, 0.268]   | 0.409               |
| $\Delta\alpha$ | +0.127   | [+0.062, +0.196] | +0.254              |

0.205 at 75; 0.360 at 85). The estimated egocentric contribution therefore *followed the fitted shape* with age; per the protocol’s language rule we do not describe it as a monotonic increase. Notably, the freely fitted quadratic placed its turning point at the same age ( $\approx 38$ ) that Bradford et al. (2023) identified as the onset of decline, while the knee-shaped sensitivity model’s knee parameter was declared unidentified under the prespecified criterion (95% profile interval spanning 26 years) — the smooth model recovers the breakpoint description without being told it, and the data cannot support estimating it as a parameter.

## 4.2 Held-out participant prediction

The age-dependent model produced a positive held-out log-likelihood difference over the fitted constant- $\alpha$  model in 20 of 20 repeats (mean +9.35, median +9.56, range [+7.06, +10.90]), meeting the prespecified stability rule R2. The per-participant mean out-of-fold difference was +0.035, positive for 65.2% of participants (descriptive summaries; no  $p$ -value is attached, as repeats are correlated). The overall outcome category is "Meets prespecified robustness criteria." The bootstrap completed 1000/1000 refits without failures.

## 4.3 A prespecified model failure

T1 passed: the observed pooled egocentric rate at ages 73–86 (0.290) fell within the model’s 95% predictive interval [0.156, 0.358]. T2 failed beyond the 99% interval: 18 participants aged 20–37 made exactly one egocentric error, against a predictive interval of [1, 11]. The fitted hierarchy, which accounts for the age trend and for heavy individual heterogeneity ( $\kappa \approx 1.4$ ), cannot reproduce the observed excess of single-error young adults. We report this as a result: a single age-varying mixture process, with this response rule, does not fully account for errors at both ends of the adult



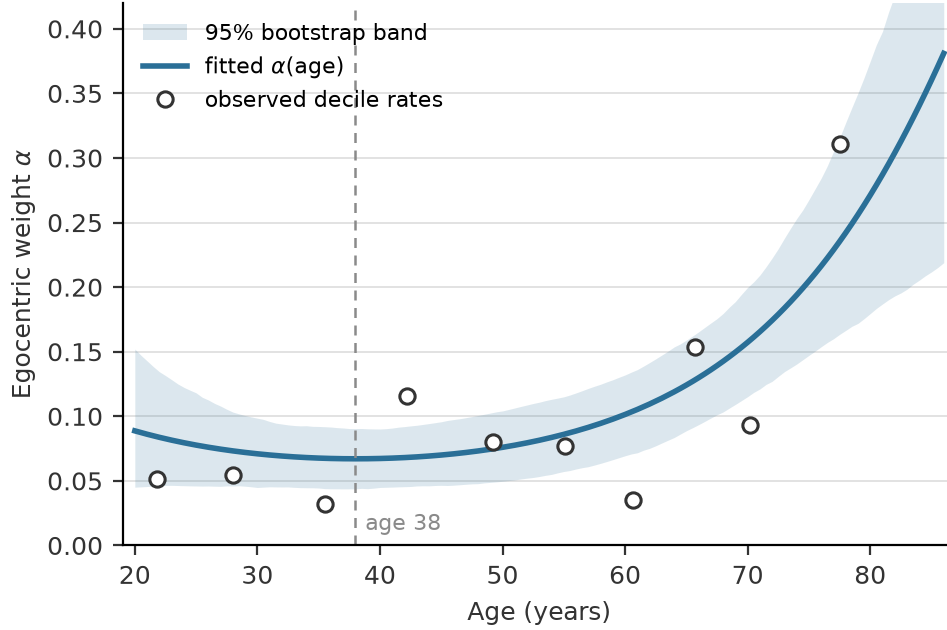


Figure 1: Fitted egocentric weight  $\alpha(\text{age})$  (line) with its 95% band from 1,000 participant-level bootstrap resamples (deterministic replay of the protocol’s seeded bootstrap), observed pooled egocentric rates by age decile (open circles;  $\approx 26$  participants each), and the fitted minimum at age 38.0 (dashed line). All values are conditional on the probability-matching response rule; under the argmax-plus-lapse reading every value doubles ( $\varepsilon = 2\alpha$ ; Proposition 1).

age range.

#### 4.4 Sensitivity analyses

The estimand was stable in every prespecified probe: fine grids,  $\Delta\alpha = +0.1270$  (unchanged to four decimals); IQ cutoffs from none to 80,  $\Delta\alpha$  between  $+0.1225$  and  $+0.1270$ ; full leave-one-participant-out, maximum  $|\Delta\Delta\alpha| = 0.0099$  ( $\approx 8\%$  of the estimate, below the prespecified flag); leave-one-trial-position-out, maximum  $|\Delta\Delta\alpha| = 0.0100$  (with the recorded caveat that trial position may conflate item identity in this dataset).

## 5 Discussion

The present analysis met the prespecified robustness criteria. Under P-MIX and its response assumptions, the estimated egocentric contribution followed the fitted age shape, with a model-implied contrast of  $\Delta\alpha = +0.127$  between ages 25 and 75 and a 95% participant-bootstrap interval of

[+0.062, +0.196]. The age-dependent model also predicted held-out participants completing the same task and item set better than the fitted constant- $\alpha$  model in all 20 cross-validation repetitions. At the same time, a prespecified predictive check failed at the younger end of the age range, showing that the fitted hierarchy did not reproduce the observed excess of participants who made exactly one egocentric error. Taken together, the results support the predictive usefulness of an age-varying account while also identifying a clear boundary on what that account can explain.

The parameter  $\alpha$  should not be interpreted as a directly measured psychological substance or as a uniquely identified degree of egocentrism. In the present projection,  $\alpha$  is a model-defined quantity: under the probability-matching rule, it indexes the probability assigned to the egocentric alternative on a critical two-choice trial. The same choice likelihoods can be generated by an argmax-plus-lapse process in which the common-ground target is normally selected but occasional random lapses produce competitor choices, with the exact mapping  $\varepsilon = 2\alpha$ . In plain terms, a fitted  $\alpha$  of .10 can be read either as 10% weight on the egocentric alternative under probability matching or as a 20% uniform lapse rate under the alternative response rule. Final choices alone cannot determine which reading is psychologically correct, so every estimate reported here remains conditional on the response process assumed by the model.

The fitted quadratic reached its minimum at approximately age 38, closely matching the point at which Bradford et al. (2023) described the onset of increasing egocentric errors in their accuracy analysis. This convergence is notable because the primary model was not given a fixed breakpoint at 38; the turning point emerged from the smooth age function. However, the estimated-knee sensitivity model did not identify a sufficiently precise breakpoint, with a profile interval spanning 26 years under the prespecified criterion. These findings should therefore not be interpreted as evidence that perspective taking changes at a sharply defined psychological threshold of exactly 38 years. Rather, the data support a smooth description in which the fitted trajectory is relatively flat through earlier adulthood and rises more strongly thereafter, while remaining too weak to distinguish that smooth pattern from a broad range of possible breakpoint locations. The agreement with Bradford et al. (2023) is therefore best treated as descriptive convergence, not independent confirmation of an exact onset age.

The most informative limitation was the failure of the young-adult singleton check. Eighteen participants aged 20–37 made exactly one egocentric error, whereas the model’s 99% predictive in-

terval extended only from 1 to 11. The fitted Beta–Binomial hierarchy can accommodate substantial stable heterogeneity between participants, but it does not generate enough isolated one-off errors among otherwise highly accurate younger adults. This discrepancy suggests that one age-varying mixture process is insufficient to reproduce the full distribution of behavior. One possible elaboration would add a separate low-frequency lapse component on top of participant-level perspective weighting. Another would allow participants or trials to arise from a mixture of processes, such that some errors reflect relatively stable egocentric influence while others reflect transient attentional failure, misunderstanding, or item-specific disruption. Neither account was fitted here, and the present result does not establish which explanation is correct. Its value is narrower but substantial: the model’s failure is localized, quantified, and capable of generating new discriminating predictions.

Several limits constrain the scope of the conclusions. First, participant-level cross-validation demonstrates prediction of unseen participants only within the observed director task and item set; it does not establish generalization to new stimuli, task variants, populations, laboratories, or communicative settings. Second, the model operates at the level of final choices and therefore discards the temporal dynamics that could separate early integration, serial correction, persistent competition, and isolated lapses. Third, the age pattern is cross-sectional, so it cannot establish within-person developmental change or a causal effect of aging. Fourth, the analysis reuses one published dataset rather than testing an independent sample. Finally, the frozen protocol followed substantial exploration of the same data, including discovery of the  $\kappa$ -floor problem and the predictive-check targets. The final execution therefore provides prespecified robustness evidence, not an independent preregistered confirmation.

The next analysis is designed around these limits rather than around extending the present headline. The source dataset contains eye-tracking outcomes that have not yet been inspected for the present project, creating an opportunity for a genuinely prospective test if the analysis protocol is frozen before those outcomes are examined. Such process-level data could distinguish whether isolated young-adult errors are preceded by sustained attraction to the privileged competitor, delayed correction toward the target, or little systematic competitor interference at all. By contrast, analyses of executive-function measures will be governed by a separate frozen protocol and labeled exploratory, as already required by the failure of the prespecified young-adult predictive check. These two next steps serve different purposes: eye tracking can test process-level predictions,

whereas executive-function analyses can examine correlates of individual variation without being treated as independent confirmation of the present age effect.

More broadly, this study shows what can be gained by placing a small explicit model between a published behavioral pattern and its psychological interpretation. The age curve of Bradford et al. (2023) described when egocentric errors became more common. The present reanalysis translated that curve into an estimated model quantity, evaluated whether the age-dependent account predicted held-out participants, and identified a pattern the model could not reproduce. The result is not a uniquely identified mechanism. It is something more modest and more useful: a transparent quantitative account with an empirical success criterion and a testable failure mode.

## Data and Code Availability

All analysis code, the frozen protocol with dated addenda, and the verbatim execution log are public at <https://github.com/francislinlfh-lgtm/Aspire-Project> (key commits: freeze e09d306; amendments a8de96b, be30e90; implementation 63943b1; results 9764283). The dataset is Bradford et al. (2023)’s open data at <https://osf.io/2epsu/>, cited rather than redistributed.

## AI Assistance Disclosure

This project was conducted by the author with substantial assistance from Claude (Anthropic). The author developed the research question, selected the theoretical model, made the substantive modeling and analysis decisions, directed and froze the robustness protocol, verified the reconstructed dataset against its published anchors, read and verified the core sources, interpreted the results, and takes full responsibility for all claims and conclusions. Claude served as a research and engineering assistant: it implemented and debugged the analysis pipeline and figure code, drafted the protocol document and the Model, Method, and Results sections for the author’s verification, performed literature searches whose key sources the author read directly, and provided methodological critique — including two rounds of structured adversarial review, elicited by the author in sessions separate from the drafting context, that produced the protocol’s amendments. All numerical results were produced by the publicly committed analysis pipeline and checked against the preserved execution

logs; the repository’s commit history records this division of labor in detail. The Introduction, Discussion, and Abstract were written by the author, with AI feedback during revision.

## Acknowledgments

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## A Protocol deviations and addenda

The analysis protocol was frozen in the public repository before the outcome-bearing execution and amended twice, both times before any execution code existed. The verbatim protocol and its full amendment history are version-controlled at <https://github.com/francislinlfh-lgtm/Aspire-Project> (file PREREGISTRATION.md); this appendix summarizes the record.

**Freeze.** Commit e09d306 locked the cleaning rules with expected values, the primary model and estimand, the estimation procedure, the cross-validation design and robustness criteria, the bootstrap, the predictive checks, and the sensitivity analyses.

**Addendum 1 (commit a8de96b; following elicited adversarial review — see the AI Assistance Disclosure for its provenance).** Ten amendments, the substantive ones being: the exercise was reclassified as a frozen robustness analysis following exploration, with outcome categories renamed accordingly; an invalid binomial  $p$ -value attached to correlated cross-validation repeats was removed, leaving the 17-of-20 threshold as a descriptive stability rule; “posterior predictive” terminology was corrected to “parametric predictive”; the cross-validation scope was restricted to unseen participants on the same task and items; the assumptions licensing the Beta–Binomial count likelihood were stated explicitly; item-position diagnostics were added; monotonicity and turning-point reporting rules were locked; the estimator was upgraded from a lattice to grid-initialized continuous optimization with synthetic-only validation; the bootstrap was fully specified and enlarged to  $B = 1000$ ; and executive-function analyses were removed to a separate future protocol.

**Addendum 2 (commit be30e90; audit corrections before execution).** The freeze and Addendum 1 had initially been labeled with dates that had not yet occurred (an authoring error; the machine clock was verified correct); all dates were corrected to the authoritative commit timestamps. Ambiguous grid-resolution language was replaced with exact step sizes, a typo was corrected, and “single execution” was defined as one outcome-bearing execution under the implementation, with any later-discovered defect requiring a dated addendum, a correction, and a complete rerun rather than preservation of a known-wrong output.

**Execution.** The implementation (commit 63943b1) was validated exclusively against synthetic data with known parameters and then executed once on the real dataset; the verbatim log is committed (`results/exp4_protocol_output.txt`, commit 9764283). No deviations from the amended protocol occurred during or after execution, and no addenda postdate the execution.